

ABDULLAH KHAN

AI Engineer • Full-Stack & Agentic AI

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Computer Science undergraduate and AI/Full-Stack Engineer building end-to-end AI-powered applications: LLM orchestration, RAG and multi-agent systems, backend services, databases, and user-facing apps. Experienced across Python, FastAPI, Django, React/Next.js, Flutter, PostgreSQL, and agentic AI frameworks (LangChain, LangGraph, CrewAI, AutoGen, MCP). IBM RAG & Agentic AI certified; currently building ARIC, an ambient multi-agent AI system, as a Final Year Project.

CERTIFICATIONS

- IBM RAG and Agentic AI Professional Certificate — 10-course specialization: RAG applications, vector databases & retrievers, multimodal generative AI, AI agent fundamentals, agentic AI with LangChain/LangGraph, multi-agent systems with CrewAI/AutoGen/BeeAI, and building agents with MCP. [Verify](#)
- Mathematics for Machine Learning and Data Science Specialization (DeepLearning.AI) — linear algebra, calculus and probability/statistics foundations for ML. [Verify](#)
- Mobile App Development Certificate — NAVTTC Prime Minister's Youth Development Program (PMYDP), Batch 3, delivered by Uptech Solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Frontend & Mobile: React, Next.js, Redux, TypeScript, Flutter

Backend & Data: Django, FastAPI, Node.js, Express.js, Supabase (Edge Functions, Auth, Row-Level Security), PostgreSQL, Redis, Celery, REST APIs

AI / LLM: LLM API integration, prompt engineering, RAG, vector databases & retrievers, function/tool calling, multi-agent systems (LangChain, LangGraph, CrewAI, AutoGen), MCP (Model Context Protocol)

Tools: Git/GitHub, Vercel, Vite

EXPERIENCE

Tekroniks — Co-Founder & Full-Stack Engineer | *2-person software agency* Dec 2025 – Present

- Co-founded a small software agency; served as the sole full-stack engineer while the co-founder led business operations and client acquisition.
- Delivered the agency's first client engagement: a Django-backed ERP (Flutter frontend, PostgreSQL) for Al Noor Cloth House (Jan – May 2026).
- Built a React site for a second client, The Wool House, completing a V1 [\(the-wool-house.vercel.app\)](#); paused after Phase 1.
- Designed and built the agency's own website (React, Redux) and its in-house SaaS product, ClassMind (see Projects below).

PROJECTS

ARIC — Ambient Real-time Intelligence Companion | *Final Year Project, In Progress — Python, FastAPI, Flutter, PostgreSQL + pgvector, Redis, Celery, WebSockets, MCP* 2025 – Present

- Designing a voice-first, cross-platform ambient AI assistant that persists memory via PostgreSQL/pgvector, routes tasks across multiple LLM providers, and executes actions across connected tools (Notion, Gmail, Slack, MCP-compatible services) with human-in-the-loop approval.
- Designed a pre-execution brainstorm/refinement engine that filters and validates action plans before any LLM-driven action is taken, reducing scope creep and hallucinated execution.
- Architecting a FastAPI microservices backend with Celery + Redis for background jobs and WebSocket-based real-time sync; MVP milestone targets an end-to-end voice-triggered action loop. Supervised by Dr. Zohaib Ahmed, NUML.

ClassMind — AI-Assisted Academic Administration Platform | *Tekroniks in-house SaaS — Flutter, Supabase (PostgreSQL, Auth, RLS, Edge Functions, TypeScript)*

- Architected an academic administration platform on Supabase/PostgreSQL with a Flutter frontend, designing the database schema, authentication and Row-Level Security policies for role-based access across student records and administrative workflows.
- Built an AI-powered Supabase Edge Function (TypeScript) that integrates an LLM to parse unstructured Excel files and extract structured student data directly into the database in a single click, eliminating manual data entry.

iLearn — AI-Powered Student Performance Assistant | *Python, FastAPI, React, Next.js, TypeScript, LLM Integration, REST APIs*

- Built an LLM-powered assistant with a FastAPI backend [\(source\)](#) and a React/Next.js (TypeScript) frontend that consumes student performance data via a JSON REST API and translates structured metrics into personalized, natural-language insights and recommendations.

EDUCATION

BS Computer Science | *National University of Modern Languages (NUML), Islamabad* 2024 – 2027 (Expected)